

# CENK CELIK

## Research Fellow, University College London

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Research Fellow with expertise in independent project leadership and a strong ethos of transparency, integrity and collaboration. Computational biologist specialising in cellular state transitions and tumour microenvironments, focusing on cell cycle arrest and EMT in cancer. Skilled in spatial and single-cell transcriptomics to dissect tumour heterogeneity and immune dynamics. Trained across institutions in Singapore and the UK, with combined wet lab and computational experience. Strong track record in high-impact publications, tool development and postgraduate supervision. Committed to advancing translational cancer research through interdisciplinary and innovative approaches.

## Professional Experience

**Research Fellow** - University College London, United Kingdom

Jun 2023 – present

- Investigating G0 arrest in breast cancer using bulk, single-cell and spatial transcriptomics datasets
- Researching cell states such as EMT in high dimensional datasets
- Developing computational methods for spatial transcriptomics data
- Mentoring master's and PhD rotation students, with weekly meetings one-on-one

**Research Fellow** - Nanyang Technological University, Singapore

Mar 2021 – May 2023

- Led a project researching *in vivo* and *in vitro* host-pathogen synergy in wound infection
- Conducted and analysed next-generation sequencing (bulk & single-cell RNA-seq) data by developing pipelines in *R* and *Python*
- Supported and supervised undergraduate and PhD students
- Utilised *C. elegans* and mouse disease models for different projects focusing on pathologies of endoplasmic reticulum stress
- Drafted, edited, revised and contributed data to papers for submission in high-impact journals
- Trained new research staff

**Research Assistant** - National University Hospital System, Singapore

Nov 2019 – Aug 2020

- Produced and analysed data to create representative graphs highlighting results for presentations
- Planned, modified and executed research techniques, procedures and tests for nano-fibrous scaffold fabrication
- Performed statistical, qualitative and quantitative analysis
- Mentored new research staff

**Graduate Assistant** - National University of Singapore, Singapore

Jan 2016 – May 2018

- Instructed undergraduate students on the fundamentals of cell culture as well as methods for collecting, analysing and reporting data
- Mentored master's students throughout all phases of their research projects

## Education

**PhD in Medicine** - National University of Singapore, Singapore

Aug 2015 – Aug 2020

- Researched scaffold-mediated stem cell chondrogenesis by pulsed electromagnetic fields
- Published a first-author research article (IF = 10.633) and contributed to another research article (IF = 8.079)
- Presented at two international conferences
- Awarded Singapore International Graduate Award scholarship by NUS
- Awarded Young Orthopaedic Researcher Award by APOA Sports Meeting '19
- Awarded NUS Researcher Travel Grant by the National University of Singapore
- Wrote a book chapter as a first author

**MSc in Bioengineering** - Ege University, Turkey

Jul 2012 – May 2015

- Produced and characterised hybrid tissue scaffold approaches for 3D vascularised bone tissue
- Awarded TUBITAK 1002 and TUBITAK Career Grant scholarships

**BSc in Bioengineering** - Ege University, Turkey

Sep 2007 – Jun 2012

- Received Republic of Turkey Prime Minister Scholarship (4 years)
- Awarded and administered TUBITAK 2209 University Students Research Projects Support Program

## Publications

### Research articles

- **Celik C\*** & Secrier M (2025). EnrichMap: Spatially informed enrichment analysis for functional interpretation of spatial transcriptomics. *bioRxiv*. [doi: [10.1101/2025.05.30.656960](https://doi.org/10.1101/2025.05.30.656960)]. \*corresponding author
- da Silva RAG, Tien BYQ, Hsien PNK, **Celik C**, Tan AZC, Ismail MH, Hu G, Chong KKL, Thibault G, Chen J, Kline KA (2025). *Enterococcus faecalis*-derived lactic acid facilitates persistent and polymicrobial wound infections by suppressing macrophage activation. *bioRxiv*. [doi: [10.1101/2025.01.31.635924](https://doi.org/10.1101/2025.01.31.635924)]
- **Celik C**, Withnell E, Pan S, Chu T, Labbadia J, Secrier M (2025). Balancing tumour proliferation and sustained cell cycle arrest through proteostasis remodelling drives immune niche compartmentalisation in breast cancer. *bioRxiv*. [doi: [10.1101/2025.01.08.632014](https://doi.org/10.1101/2025.01.08.632014)]
- **Celik C**, Lee SYT, Tanoto FR, Veleba M, Kline KA, Thibault G (2024). Decoding complexity of delayed wound healing following *Enterococcus faecalis* infection. *eLife* 13:RP95113. [doi: [10.7554/eLife.95113.3](https://doi.org/10.7554/eLife.95113.3)]
- Kaliya-Perumal AK, **Celik C**, Carney TJ, Harris MP, Ingham PW (2024). Genetic regulation of injury induced heterotopic ossification in adult zebrafish. *Dis Model Mech* dmm.050724. [doi: [10.1242/dmm.050724](https://doi.org/10.1242/dmm.050724)]
- Dudkevich R, Koh JH, Beaudoin-Chabot C, **Celik C**, Lebenthal-Loinger S, Karako-Lampert S, Thibault G, Henis-Korenblit S (2022). Neuronal IRE-1 coordinates an organism-wide cold-stress response by regulating fat metabolism. *Cell Reports* 41, 111739. [doi: [10.1016/j.celrep.2022.111739](https://doi.org/10.1016/j.celrep.2022.111739)]
- Beaudoin-Chabot C, Wang L, **Celik C**, Abdul-Khalid ATF, Thalappilly S, Xu S, Koh JH, Lim VWX, Low AD, Thibault G (2022). The unfolded protein response reverses the effects of glucose on lifespan in chemically-sterilised *C. elegans*. *Nat Commun* 13, 5889. [doi: [10.1038/s41467-022-33630-0](https://doi.org/10.1038/s41467-022-33630-0)]
- Yang Z, **Celik C**, Parate D, Franco-Obregón A, Lee EH (2022). The application of pulsed electromagnetic field for cartilage regeneration. *Tissue Eng Part A* 28, S-365. [doi: [10.1089/ten.tea.2022.29025.abstracts](https://doi.org/10.1089/ten.tea.2022.29025.abstracts)]
- **Celik C**, Franco-Obregón A, Hui JHP, Lee EH & Yang Z (2021). Directionalities of magnetic fields and topographic scaffolds synergise to enhance MSC chondrogenesis. *Acta Biomater* 119, 169-183. [doi: [10.1016/j.actbio.2020.10.039](https://doi.org/10.1016/j.actbio.2020.10.039)]
- Parate D, Kadir ND, **Celik C**, Lee EH, Hui JHP, Franco-Obregón A & Yang Z (2020). Pulsed electromagnetic fields potentiate the paracrine function of mesenchymal stem cells for cartilage regeneration. *Stem Cell Res & Ther* 11:46. [doi: [10.1186/s13287-020-1566-5](https://doi.org/10.1186/s13287-020-1566-5)]
- Cetmi SD, Renkler NZ, Kose A, **Celik C** & Oncel SS (2019). Preparation of electrospun polycaprolactone nanofiber mats loaded with microalgal extracts. *Eng Life Sci* 19:691-699. [doi: [10.1002/elsc.201900009](https://doi.org/10.1002/elsc.201900009)]
- Duman OM, **Celik C**, Sarikanat M & Sendemir A (2014). Investigation of central nervous system neurons under mechanical strain: An *in vitro* traumatic brain injury model. *18<sup>th</sup> National Biomedical Engineering Meeting*, 1-4. [doi: [10.1109/biomut.2014.7026384](https://doi.org/10.1109/biomut.2014.7026384)]

### Reviews

- **Celik C**, Pan S, Withnell E, Lam HW, Secrier M (2025). Decrypting cancer's spatial code: from single cells to tissue niches. *Mol Oncol* [in press].
- **Celik C**, Lee SYT, Yap WS, Thibault G (2023). Endoplasmic reticulum stress and lipids in health and disease. *Progress in Lipid Research* 8, 101198. [doi: [10.1016/j.plipres.2022.101198](https://doi.org/10.1016/j.plipres.2022.101198)]

### Book chapters

- **Celik C**, Mogal VT, Hui JHP, Loh XJ & Toh WS (2018). Injectable Hydrogels for Cartilage Regeneration. In: Thakur V, Thakur M (eds) *Hydrogels. Gels Horizons: From Science to Smart Materials*. Springer, Singapore. [doi: [10.1007/978-981-10-6077-9\\_12](https://doi.org/10.1007/978-981-10-6077-9_12)]
- Karaman O, **Celik C** & Sendemir A (2018). Self-Assembled Biomimetic Scaffolds for Bone Tissue Engineering. In Management Association, I. (Ed.), *Biomedical Engineering: Concepts, Methodologies, Tools and Applications*, 476-504, Hershey, PA: IGI Global. [doi: [10.4018/978-1-5225-3158-6.ch021](https://doi.org/10.4018/978-1-5225-3158-6.ch021)]

## Invited talks

- **Celik C** (May 10, 2025). Spatial evolution of proliferative and arrested niches in breast cancer. *AI in Pharma*, Hacettepe University, Ankara, Türkiye.
- **Celik C** (Sep 23, 2024). Internal and external regulation of cell cycle arrest in breast cancer. *Izmir Biomedicine and Genome Centre*, Izmir, Türkiye.

## Proceedings

- **Celik C**, Withnell E, Secrier M (Oct 7, 2024). Intrinsic and extrinsic factors influencing G0 cell cycle arrest in breast cancer [Selected talk]. *Crick Cancer Research Symposium 2024*, London, UK.
- **Celik C**, Withnell E, Secrier M (May 1, 2024). Unravelling internal and external regulatory mechanisms of cell cycle arrest in breast cancer [Poster]. *UCL Cancer Symposium 2024*, London, UK.
- **Celik C**, Secrier M (Jan 24-25, 2024). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer [Selected talk & Poster]. *Festival of Genomics & Biodata*, London, UK.
- **Celik C**, Secrier M (Oct 8-10, 2023). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer: Insights from single-cell data [Poster]. *4<sup>th</sup> Crick International Cancer Conference*, London, UK.
- Lee STL, **Celik C**, Tan AMZ, Veleba M, Kline KA, Thibault G (Dec 3-7, 2022). Dissecting the synergistic role of the unfolded protein response in wound infections [Poster & Oral]. *ACSB & EMBO Meeting*, Washington DC, USA.
- **Celik C**, Yang Z, Franco-Obregón A, Hui JHP (Apr 4-7, 2019). Altering MSC chondrogenesis by PEMF [Oral]. *APOA Sports Meeting 19*, Kuala Lumpur, Malaysia.
- **Celik C**, Yang Z, Franco-Obregón A, Hui JHP (Sep 9-11, 2018). Enhancement of MSC chondrogenesis by PEMF [Poster]. *5<sup>th</sup> TERMIS World Congress*, Kyoto, Japan.
- Cetmi SD, Renkler NZ, Kose A, **Celik C**, Sendemir A & Oncel SS (Nov 11, 2015). Utilisation of microalgal extracts for construction of tissue scaffolds with electrospinning techniques [Poster]. *7<sup>th</sup> International Bioengineering Congress*, Izmir, Türkiye.
- Duman OM, Tasdemir S, Minaz MC, **Celik C** & Sendemir A (Nov 11, 2013). Fe<sub>3</sub>O<sub>4</sub> reinforced polycaprolactone nanofibrous scaffolds [Poster]. *VI<sup>th</sup> International Bioengineering Congress: Human Welfare*, Kusadasi, Türkiye.
- **Celik C**, Karaman O, Sendemir A (Nov 11, 2013). More than a picture: Biomedical Illustration [Oral]. *19<sup>th</sup> International Biomedical Science and Technology Symposium*, Kusadasi, Türkiye.
- **Celik C**, Gorgun C, Sendemir A (Nov 11, 2013). Cell viability of keratinocytes on electrospun Spirulina/PCL composites [Poster]. *19<sup>th</sup> International Biomedical Science and Technology Symposium*, Kusadasi, Türkiye.
- Duman OM, Cogan S, Minaz MC, **Celik C** & Sendemir A (Sep 9, 2013). PCL/Fe<sub>3</sub>O<sub>4</sub> Scaffold Production with Electrospinning [Oral]. *Advanced Materials World Congress 2013*, Cesme, Türkiye.
- Serdengeçti C, Sen I, Duman OM, **Celik C**, Sendemir A & Seki Y (Apr 4, 2013). Production and analysis of parameters affect morphological characters of chitosan nanospheres [Poster]. *1<sup>st</sup> Ege University Nanotechnology Days*, Izmir, Türkiye.
- Duman OM, **Celik C**, Minaz MC, Sendemir A (Apr 4, 2013). Fabrication of PEO/Fe<sub>3</sub>O<sub>4</sub> composite tissue scaffolds by electrospinning [Poster]. *1<sup>st</sup> Ege University Nanotechnology Days*, Izmir, Türkiye.
- Demirkaya C, **Celik C**, Sarikanat M, Conk-Dalay M, Sendemir A (May 5, 2012). Applications of microalgae for tissue engineering [Oral]. *Marine Biotechnology and Genomics Workshop*, Bodrum, Türkiye.
- **Celik C**, Demirkaya C, Sarikanat M, Conk-Dalay M & Sendemir A (Nov 11, 2011). Effects of microalgae reinforcement on biocompatibility and mechanical properties of tissue engineering scaffolds [Oral]. *1<sup>st</sup> National Aegean Composite Materials Symposium*, Selcuk, Türkiye.

## Teaching

- Life at the single-cell and spatial scale. *Bootcamp* (Aug 2025) at Bilimler Köyü, Foça, Türkiye.
- Art of ATGC: Genetic data analysis and visualisation. *Bootcamp* (Aug 2024) at Bilimler Köyü, Foça, Türkiye.
- Graduate Teaching Assistantship Programme (2016 - 2018). National University of Singapore.

## Proficiencies

- *R/Studio* • Python 3 • *UNIX/Bash* • Cell Ranger • Loupe Browser • NGS data analysis (bulk, single-cell RNA-seq & spatial transcriptomics)

• Documentation/Version control (GitHub, .Rmd, ReadTheDocs) • Image processing • Affinity Designer/Photo

## Core Competencies

• Single-cell & bulk RNA-seq • qPCR • Western blot • Cell culture • Confocal imaging • Immunofluorescence  
• Handling of mice & rats • Lentiviral transfection/transduction • Primer design • Single-cell isolation • Flow cytometry

## Memberships

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|----------------------------------------------------------------|---------------------|
| - <b>The European Association for Cancer Research</b> - Member | Feb 2024 – present  |
| - <b>The Genetics Society UK</b> - Member                      | Aug 2023 – Aug 2024 |
| - <b>Society for Cell Biology Singapore</b> - Member           | Sep 2022 – May 2023 |
| - <b>DIE at NTU School of Biological Sciences</b> - Member     | Mar 2022 – May 2023 |

## Certifications

- **Responsible Care & Use of Laboratory Animals** - SingHealth, Apr 2021, ID: SEMC/RCULAC/2021/040  
- **Genomic Data Science Specialisation** - John Hopkins University, Sep 2020, ID: D7S7D536DXDD  
- **Data Analysis for Life Science** - HarvardX, Aug 2020, ID: a1cb1feb86a24a6b8d28873249cf1bad  
- **Python Data Science** - edX & IBM, Jun 2020, ID: 8e35eee7e19b4071a2a1109e599b8800

## Awards & Honours

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|--------------------------------------------------------------------------------------------------|---------------------|
| - <b>Crick Cancer Symposium 2024</b> - Selected talk, London                                     | Oct 2024            |
| - <b>Poster Presentation Winner</b> - Festival of Genomics & Biodata, London                     | Jan 2024            |
| - <b>Translational Researcher Programme</b> - 10X Genomics, Inc., Singapore                      | Feb 2022            |
| - <b>Young Orthopaedic Researcher</b> - APOA Sports Meeting '19, Kuala Lumpur                    | Apr 2019            |
| - <b>Researcher Travel Grant</b> - National University of Singapore, Singapore                   | Sep 2018            |
| - <b>Singapore International Graduate Award</b> - National University Singapore, Singapore       | Aug 2015 – Aug 2019 |
| - <b>Research Scholarship</b> - TUBITAK 1002, Turkey                                             | Jun 2014 – May 2015 |
| - <b>Research Scholarship</b> - TUBITAK Career Grant, Turkey                                     | Sep 2012 – Sep 2013 |
| - <b>University Students' Research Project Support Program</b> - TUBITAK 2209, Turkey            | Sep 2011 – Jan 2012 |
| - <b>Prime Minister Scholarship</b> - General Directorate of Credit & Dormitories Agency, Turkey | Jan 2008 – Jun 2012 |

## Volunteering

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|----------------------------------------------------------------------------------------------------------|---------------------|
| - <b>Life at the single-cell and spatial scale</b> - Bilimler Köyü                                       | 11–17 Aug 2025      |
| - <b>Computational Research Exchange Cancer Network</b> - UCL                                            | Dec 2024 – Mar 2025 |
| - <b>Art of ATGC: Genetic Data Analysis and Visualisation</b> - Bilimler Köyü                            | 12–18 Aug 2024      |
| - <b>Bioinformatics Instructor</b> - KizCode                                                             | Oct 2021 – Jan 2024 |
| - <b>Executive Committee Member</b> - Singapore-Turkey Friendship Association                            | May 2016 – Jun 2018 |
| - <b>Organising Committee</b> - 19 <sup>th</sup> International Biomedical Science & Technology Symposium | Nov 2012 – Dec 2012 |
| - <b>Organising Committee</b> - 6 <sup>th</sup> International Bioengineering Congress                    | Nov 2012 – Dec 2012 |